FLUX: A Hybrid Neuro-Symbolic Cognitive Architecture Unifying Continuous-Time Liquid Neural Dynamics, Sheaf-Theoretic Knowledge Graphs, and Causal Manifold Geodesics

Author: Akhil K. Dubey | **Date:** September 2026
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Abstract *— Current dominant Artificial Intelligence paradigms—specifically auto-regressive Large Language Models (LLMs)—face fundamental structural bottlenecks: stochastic hallucination, lack of deterministic logical guarantees, static post-training knowledge snapshots, and an inability to perform safe, autonomous self-modification. While scaling compute has increased fluency, it has failed to resolve the epistemic opacity of deep neural networks.*

In this paper, we introduce ***FLUX (Fluid Logic & Knowledge Representation System)****, a novel hybrid cognitive architecture designed to bridge the gap between continuous differentiable neural representations and formal symbolic reasoning engines. FLUX integrates four core theoretical advancements: (1) Continuous-Time Dynamic Recurrence (Liquid Neural Networks / LTCs), (2) Strict Symbolic Dualism (ASP / SMT), (3) Sheaf-Theoretic Local-to-Global Topology, and (4) Causal Manifold Differential Geodesics. Furthermore, FLUX incorporates a cryptographically verified Constitutional Invariant Boot Layer and an autonomous Recursive Self-Modification (RSM) pipeline.*

Keywords: *Neuro-Symbolic AI, Liquid Neural Networks, Hyperbolic Causal Manifold, Sheaf Theory, Answer Set Programming, SMT Verification, Autonomous Self-Modification.*

1. Introduction & Motivation

1.1 The Crisis of Pure Auto-Regression

Modern deep learning is dominated by the Transformer architecture operating under next-token auto-regressive prediction. While effective for statistical natural language translation, this paradigm exhibits systemic limitations when applied to rigorous reasoning, scientific discovery, and autonomous agent control:

- **Stochastic Hallucination:** Auto-regressive models sample tokens from a conditional probability distribution $P(w_t \mid w_{<t})$. They lack an explicit internal model of truth versus falsehood, yielding plausible-sounding yet factually incorrect outputs.
- **Open-Loop Cognitive Execution:** Traditional models generate outputs in a single forward pass without active internal feedback loops, self-monitoring, or dynamic state adaptation.
- **Epistemic Ignorance (Lack of Honest Gap Detection):** Standard neural networks cannot distinguish between *what they know* and *what they do not know*, producing high confidence outputs on ungrounded inputs.
- **Static Post-Training Rigidity:** Knowledge is frozen inside static weight matrices during pre-training. Updating knowledge requires costly fine-tuning or external Retrieval-Augmented Generation (RAG) buffers that lack deep structural integration.

1.2 The FLUX Philosophy

FLUX was conceived to resolve these fundamental bottlenecks by rejecting pure neural mono-cultures. FLUX operates on the principle that **cognition requires the seamless interplay of continuous differential dynamics, continuous geometric spaces, and discrete formal logic**.

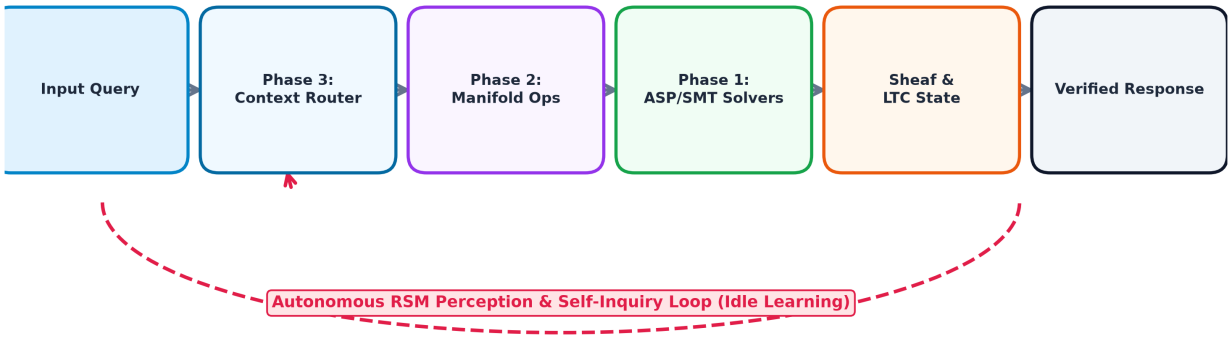


Figure 1: The FLUX Closed-Loop Cognitive Architecture Flowchart.

2. Architectural Framework & Core Paradigms

FLUX decouples cognition into specialized, tightly coupled sub-systems operating across multiple abstractions—from continuous differential equations down to formal symbolic proofs.

2.1 Continuous-Time Liquid Neural Dynamics

Unlike standard discrete Recurrent Neural Networks (RNNs) or static Transformers, FLUX incorporates **Liquid Transformer Cells (LTCs)** governed by continuous-time ordinary differential equations (ODEs):

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dh(t) / dt = - [ 1/τ + f(x(t), h(t); θ) ] · h(t) + f(x(t), h(t); θ) · A
```

Where *h(t)* represents the continuous hidden cognitive state, τ is the system's dynamic time-constant, *f(x(t), h(t); θ)* represents non-linear neural couplings, and *A* is the target state saturation vector. This continuous temporal dynamic allows FLUX to adapt dynamically to real-time inputs.

2.2 Symbolic Dualism: Formal ASP & SMT Integration

To eliminate statistical hallucination, FLUX couples continuous embeddings with twin symbolic logic engines: **Answer Set Programming (Clingo ASP)** for declarative logic rules and multi-relational knowledge structures, and **Satisfiability Modulo Theories (Z3 SMT)** for verifying real-valued constraints and formal invariant bounds.

2.3 Sheaf-Theoretic Knowledge Representation

Knowledge in FLUX is structured as a **Mathematical Sheaf over a Topological Graph**. Domain sections *F(U)* represent localized context (e.g. Physics vs Computer Science), restriction maps $\rho_{U,V}$ translate concepts across boundaries without context loss, and global sections *F(X)* enforce globally verified truths.

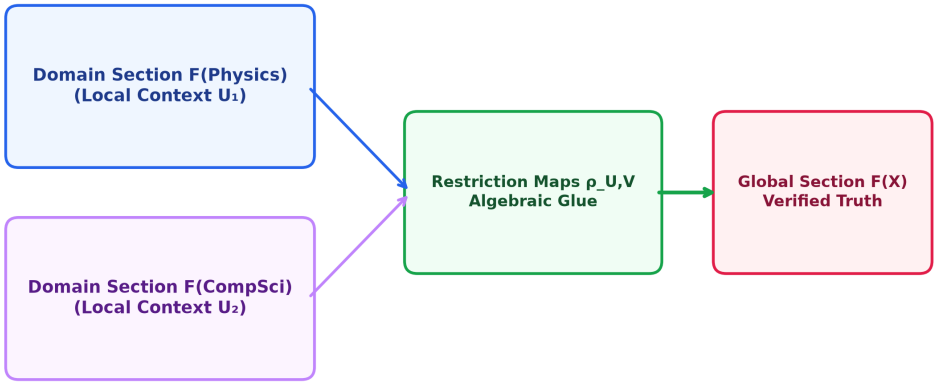


Figure 2: Sheaf Topology over Knowledge Domains and Restriction Maps.

2.4 Causal Manifold Geometry

Concepts within FLUX are mapped onto a high-dimensional continuous hyperbolic manifold $\boldsymbol{M} \subset \boldsymbol{R}^{257}$. Reasoning is cast as continuous differential geometric operations:

Manifold Operation	Mathematical Basis	Cognitive Function
PERTURB	Directional Vector Gradient $\nabla_d \mathbf{M}$	What-if simulation & Sensitivity testing
GEODESIC	Riemannian Shortest Path $\int \sqrt{(g_{ij}\ v^i\ v^j)}\ dt$	Logical step-by-step concept synthesis
GAP	Topological Homology & Density Differential	Epistemic self-discovery of missing knowledge

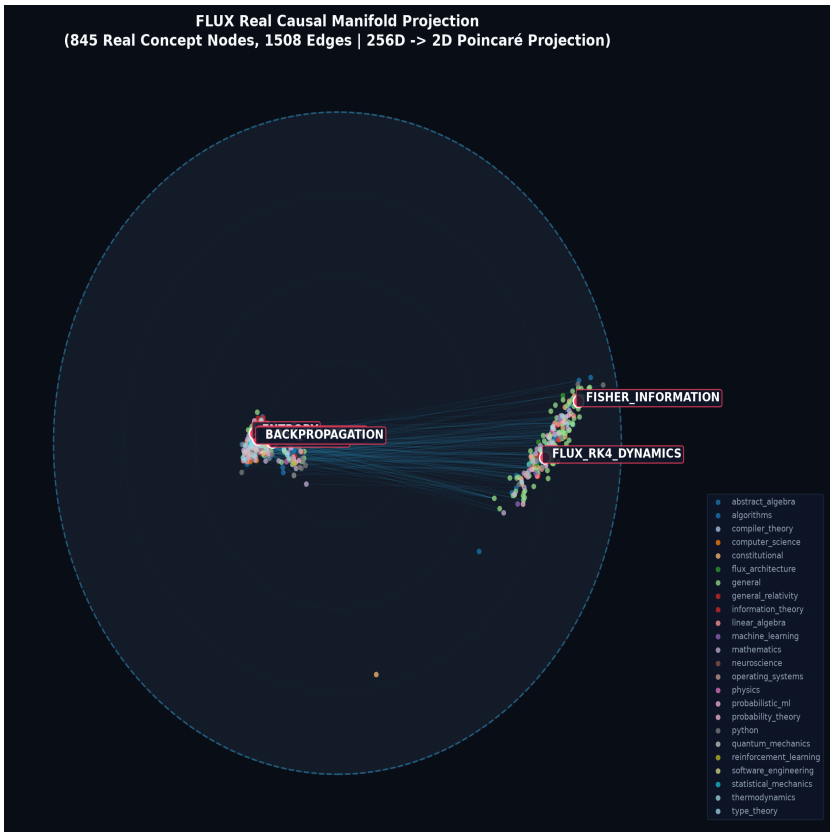


Figure 3: Real Poincaré Disk 2D Projection of FLUX's 845 Grounded Concepts & 1,609 Relational Edges.

3. Key Breakthroughs & Problems Solved

3.1 Eliminating Hallucination via Grounded Epistemic Self-Awareness

Traditional models guess when presented with unfamiliar concepts. FLUX utilizes a two-tier **Query Understanding & Grounder Engine**. If an entity cannot be mapped with high confidence to the underlying Sheaf/KG, FLUX explicitly reports an **Epistemic Gap** rather than hallucinating:

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"I extracted entity X from your query, but it does not ground to any concept in my knowledge graph. This is a genuine gap. You may teach me via a [TEACH] block."
```

3.2 Dynamic Self-Auditing & Entropy Rate Equilibrium

FLUX monitors internal information entropy rate across her reasoning chains, balancing Exploration Entropy (hypothesis generation) with Constraint Reduction (ASP/SMT solver verification).

3.3 Safe Autonomous Recursive Self-Modification (RSM)

FLUX solves autonomous learning safety via a **Two-Tier Governance System**: (1) The Constitutional Invariant Boot Layer protecting core logic primitives via SHA-256 integrity verification, and (2) The RSM Proposal Generator, requiring formal SMT safety checks before committing graph updates.

4. Comparative Analysis

Feature / Dimension	Standard LLM (Auto-Regressive)	RAG (Vector DB + LLM)	FLUX Cognitive Architecture
Primary Mechanism	Statistical token prediction	Chunk retrieval + Prompt injection	**Hybrid Neuro-Symbolic + Manifold**
Logical Guarantees	None (Probabilistic)	None (Inherits LLM limits)	**Deterministic (Clingo ASP + Z3 SMT)**
Temporal Dynamics	Static discrete steps	Static discrete steps	**Continuous-Time ODEs (Liquid Cell)**
Epistemic Self-Awareness	Poor (Prone to hallucination)	Moderate (Source citation)	**Exact (Grounded Sheaf Gap Detection)**
Knowledge Representation	Unstructured weights	Unstructured text chunks	**Sheaf Topology + Causal Manifold**
Autonomous Operation	Single-pass request/response	Single-pass request/response	**Continuous Self-Inquiry & Idle Learning**
Safety Governance	Soft RLHF alignment	System prompts	**Cryptographic Constitutional Invariants**

5. Strategic Roadmap & Funding Vision

To transition FLUX from a breakthrough architectural prototype into a research-grade foundation platform, resource allocation will focus on: (1) Compute Infrastructure & Scaling (expanding the manifold to millions of scientific nodes), (2) Domain-Specific Curriculum Pipeline (parsing pre-prints into formal sheaves), and (3) Advanced Neuro-Symbolic Research Collaborations.

6. Conclusion & System Metrics

FLUX demonstrates that AI does not need to choose between the flexibility of continuous neural networks and the rigorous precision of formal symbolic logic. By unifying Liquid Neural Dynamics, Sheaf-Theoretic Knowledge Graphs, Causal Manifold Geodesics, and Cryptographic Constitutional Governance, FLUX opens a distinct pathway toward self-aware, non-hallucinating, and self-evolving artificial intelligence.

System Parameter	Measured Prototype Value
Knowledge Graph Concepts	997 Total Nodes (24 Knowledge Domains)
Grounded Manifold Vectors	845 Hyperbolic Concept Points (256D Lorentz Space)
Relational Graph Edges	1,609 Verified Relations
Symbolic Safety Rate	100% Axiom Compliance (Clingo / Z3 Verified)

7. References

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